FALCON AUTOTECH

FALCON's Response to Delhivery's RFQ for Faridabad Warehouse Automation for Parcel Sortation

May 3, 2023

Kind Attention

Mr Sunny Raja
Mr Himanshu Sekhar
Delhivery

Offer Ref: F23-00256-00; Date 03.05.2023.
Subject – Techno-commercial Offer for the Cross-Belt Sorter_12k for Faridabad Site

Dear Team,

We are pleased to submit our Techno-commercial offer in response to your RFQ.

As you will note, we have studied your requirement in great depth and, along with the information collected during calls with you, we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposal and to re-enforce our commitment to be your partner in this important project.

In subsequent sections, we have highlighted the capabilities and experiences of Falcon Autotech with sections on our Sortation Technologies and references.

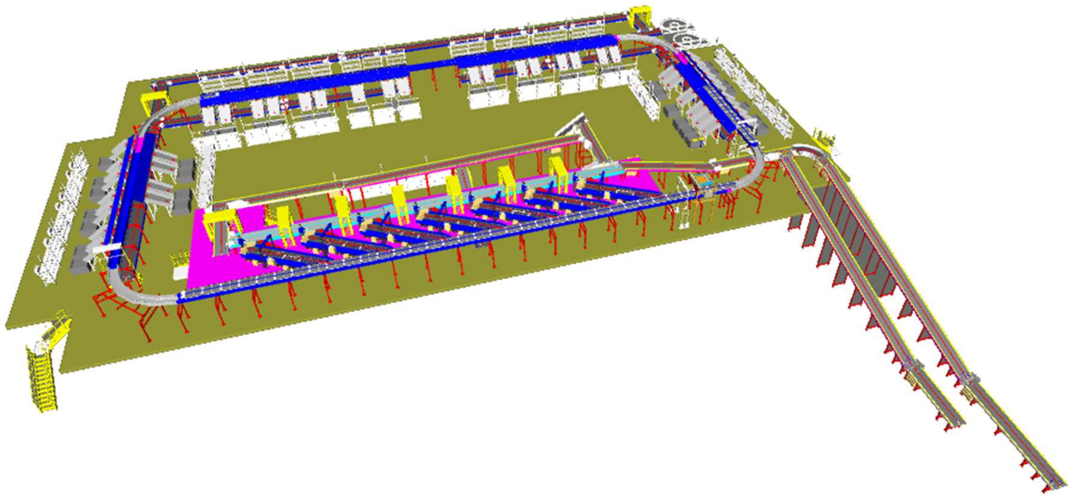
To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through your award process, please do not hesitate to contact me and my team. We will be pleased to assist you for any further information or clarifications that you might have.

Best Regards,

Adib Hasim

Manager – Sales

Response to RFP for Cross Belt Sorter_12K_Delhivery_Faridabad



Proposal Reference - F23-00256-00

Falcon Autotech Private Limited
Plot No 308 & 309
Ecotech-1 Extension, Greater Noida
Uttar Pradesh- 201308
India.

Contact – Adib Hasim
Sales Manager
Mob - +91 9711619998
Adib.hasim@falconautoonine.com

Table of Contents

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	PPH	Parcels Par Hour
3	ARB	Activated Roller Belts
4	DBO	Damaged Barcode
5	VDS	Volume Distribution System
6	ICR	Intelligent Character Recognition
7	MEZZ	Mezzanine
8	LIM	Linear Induction Motors
9	ECDS	Empty Carrier Detection System
10	AC	Alternating Current
11	DC	Direct Current
12	PLC	Programmable Logic Controller
13	IT	Information Technology
14	BOQ	Bill Of Quantity
15	I/O	Input/ Output
16	PDP	Power Distribution Panel
17	PC	Personal Computer
18	UPS	Uninterrupted Power Supply
19	CBS	Cross Belt Sorter
20	DNF	Data Not Found
21	LGM	Logic Mismatch
22	RTVC	Real Time Video Coding

2. Handled Parcel Spectrum

2.1 Parcel size loadable on the sorter

Max Length	mm	500
Max Width	mm	500
Max Height	mm	500
Max Weight	Kg	10
Min length	mm	100
Min Width	mm	50
Min Height	mm	5
Min Weight	Kg	0.02
Dimension Accuracy	mm	(+-10)
Weight Accuracy	Kg	(+-0.06)

Note: Min measurable size by dimension scanner will be 100x50x20 (mm).

2.2 Parcels to be loaded on Sorter shall have the following characteristics:

1. Centre of Gravity of item must not move during conveyance or sorting.
2. Item must not have magnetic content, otherwise behavior of parcel cannot be guaranteed.
3. Liquid or fragile material, to avoid breaking, spillage or leakage, such as wine bottles, metal cans of paint are designated as non- conveyable items.
4. Parcels shall be perfectly and safely packaged: protrusion or open surfaces are not allowed.
5. Plastic ropes shall be perfectly adherent to the surface of the package.
6. All items with the risk of being damaged during the transport on an automatic sorting system or damaging the sorting system. They must be robust enough to avoid disintegration of container material and loose of contents in the sorting process.
7. Item packaging shall have enough grip to be handled on the belts during the acceleration and referencing phases.
8. Items shall not have slippery surfaces and must be able to withstand acceleration of the items on the belt during the start-stop phases (accelerations up to 0.5 g shall be assured without any sliding or tumbling of the items on the belt conveyor).
9. The parcels must have at least one flat and regular surface providing enough stability during conveyance.
10. All shapes are permitted except spherical, cylindrical, or alike unstable items & shapes.
11. All usual packaging materials are permitted (including paper, carton, plastics, plastic foil, rope, tape, textile, and wood)

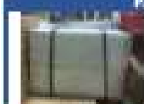
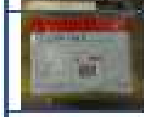
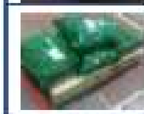
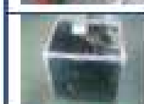
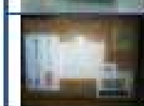
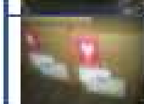
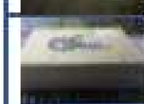
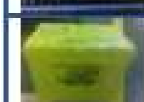
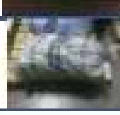
2.3 Parcel not loadable on the sorter

All products that are not within the range as described here, are considered non- conveyable products, and must be taken out of the main sorter flow by the operators.

1. Unstable items with a risk to roll or tumble on the sorting system, such as spherical or cylindrical items
2. Items that have a spherical or cylindrical shape.

3. Items that are packed in material that can damage the conveyors or the sorter.
4. Items that have sharp points (e.g. Nails) or sharp edges, that can damage the conveyors or the sorter.
5. Fragile parcels with contents not sufficiently secured
6. Items that have been classified as dangerous are designated.
7. Wet items are designated.
8. Items with anti-slip treatment.
9. Items with protruding parts.
10. Items with sharp edges.
11. Inadequately packed items that could be damaged during automatic transportation.
12. Electrostatically loaded items.
13. Loose parts on loads and load carriers, such as adhesive tape, stickers, slips of paper, straps, wrap foil etc. are designated as non- conveyable items.

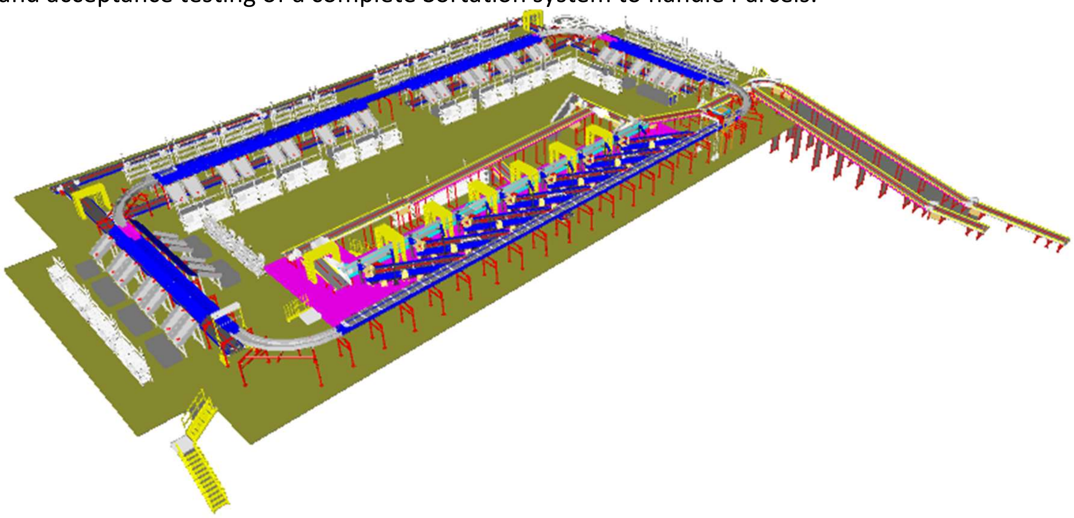
2.4 Pictures of Sortable Parcels

PARCEL TYPE		DESCRIPTION
	Strongly Strapped	A parcel banded with strapping material.
	Pre Manufactured Plastic Courier Pack <A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size.
	Pre Manufactured Plastic Courier Pack > A3 in size	A parcel shrouded in a plastic coating and is a manufactured product of a pre-determined size. Loose part with thickness under to minimum allowed (10 m) may create problems and are not recommended to be processed.
	Other Plastic Covered Item	A cardboard parcel coated in a plastic material, for example bin liner, shrink-wrap and stretch-wrap.
	Paper Covered Item	A cardboard parcel coated in a packing paper.
	Cloth Covered Item	Often import parcels that are of irregular shape.
	Wine Box	A robust, cardboard parcel that is designed to protect its fragile contents from breakage and has a high weight to size ratio.
	Cardboard Box	A cardboard parcel is often the norm.
	Totes	A large plastic, lidded box.
	Flats	A parcel that often contains documents.
	Poly Bags	It is a plasticized weave bag, used as a containerization solution, holding a number of other parcels within it.

3. Proposed System Description

3.1 Objective

The purpose of this proposal is the design, manufacturing, installation, commissioning, testing and acceptance testing of a complete Sortation system to handle Parcels.



3.2 Summary of the System

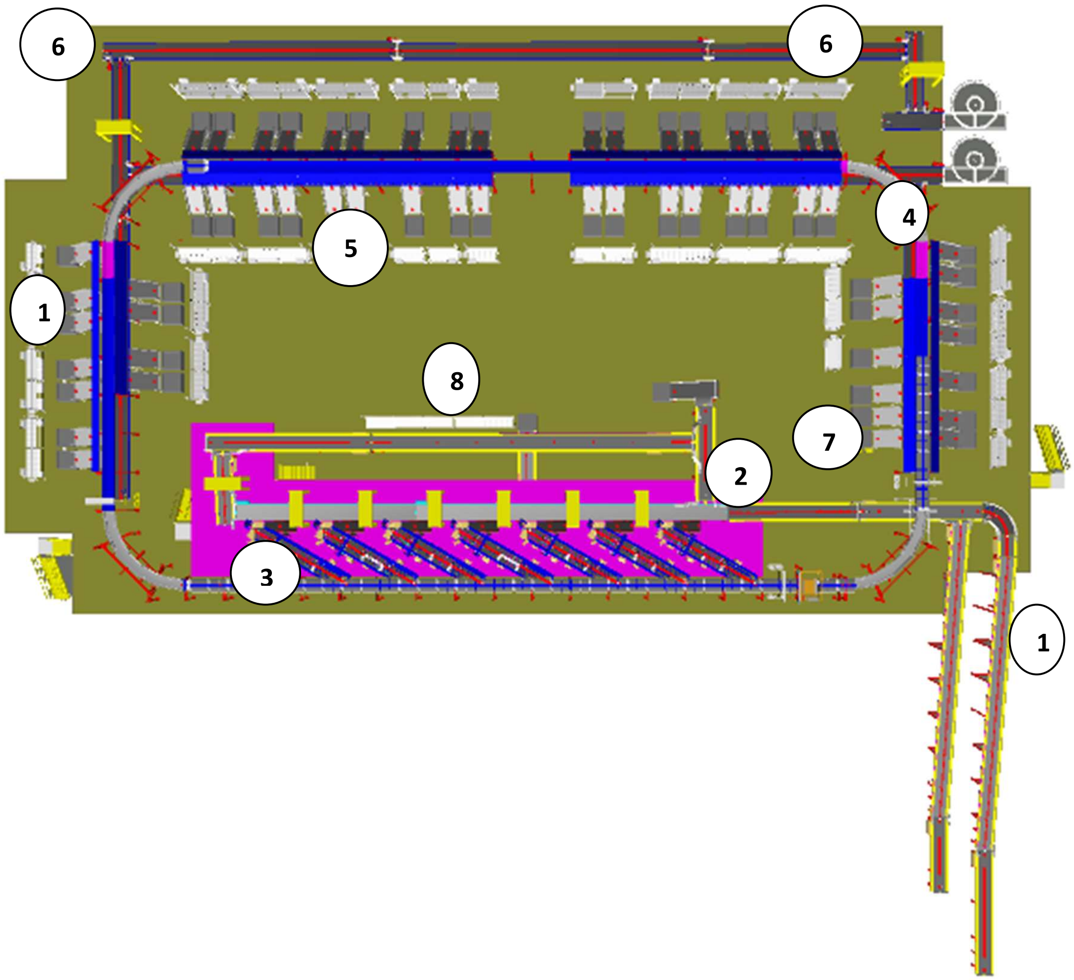
1. Bulk Infeed conveyors are used to transfer the Shipments from Ground level to Mezzanine Level to the VDS
2. Mixed Load of Parcels is dumped to the Induct Stations via loop VDS systems.
3. The VDS system feeds 7 Induct Lines.
4. Each Induct line is Equipped with a Loading Conveyor and online automatic weighing.
5. The operator picks up a parcel from the VDS conveyor and places it on the Loading Conveyor with Barcode facing upwards.
6. These Induction Lines induct the parcels on the Sorter automatically onto to the Cross-Belt Sorter which then sorts the parcels into respective chute using data from Delivery's Sorting decision system. The Dimensions and image of the parcel is also captured on the Sorter.
7. The sorted shipments are discharged via sliding/PTL chutes into Bellow type collection bins.
8. Take out Conveyors are planned below the sorter. Once the Bags are filled, they are pushed onto the Take-out Conveyors after sealing and tagging.
9. Take-away conveyors terminate in spiral chutes (Delhivery Scope) that take the bags down to the ground floor.
10. Safety devices are considered as per Falcon Standards and referenced from previous projects.
11. Control and IT Systems to suit the requirement and as per Delhivery specification.

3.3 Main benefits of the proposed solution

The proposed solution has the following advantages:

1. High operational throughput
2. Good Parcels conveying quality for the entire range of conveyable Parcels with a sorting conveyor speed up to 2.1 m/s.
3. Maximized sorting capacity throughout the entire sorter loop, including the two infeed zones.
4. Low occupancy of floor space in the building
5. Narrow discharge centers for increased number of splits in limited space
6. Scalable with increasing Business Volumes.
7. FALCON's CBS can adapt with changing business requirements by adjusting its speed, to match the operational throughput requirement, thereby leading to Power Savings and reduced system Wear & Tear

4. Layout View



Legend

1 – Infeed Line

2 – VDS Loop

3 – Feedlines

4– Main Loop

5 – PTL Chutes

6 – Takeaway Conveyors

7– Rejection Chutes

8 – non-conveyable sorting Station

5. Proposed System Capacity_ Calculations

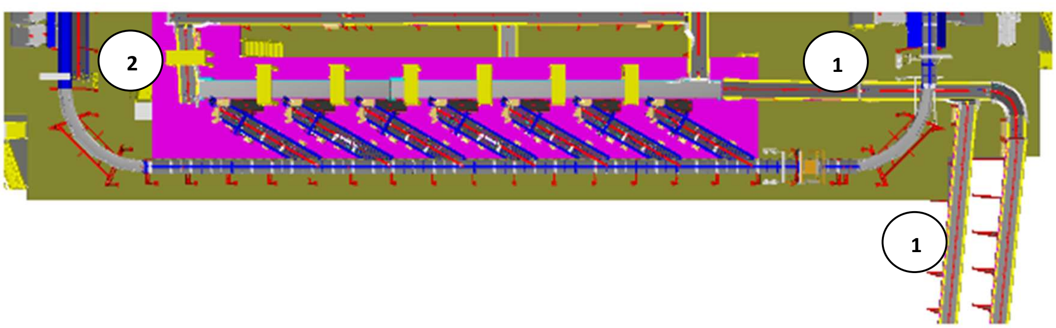
5.1 Sorter System Capacity

The following table shows the Throughput Calculation for the Sorter

Sorter Speed	2.1	m/s
Carrier Pitch (For dual belt)	1175	m
Carriers per Hour/ Zone	12600	CPH
Sorter Designed Throughput	12600	PPH
Rated TPH per Induct Line	2000	PPH
Number of Induct Lines	**7**	**QTY**

6. System description

6.1 Infeed System



Legend

1- Infeed Conveyor

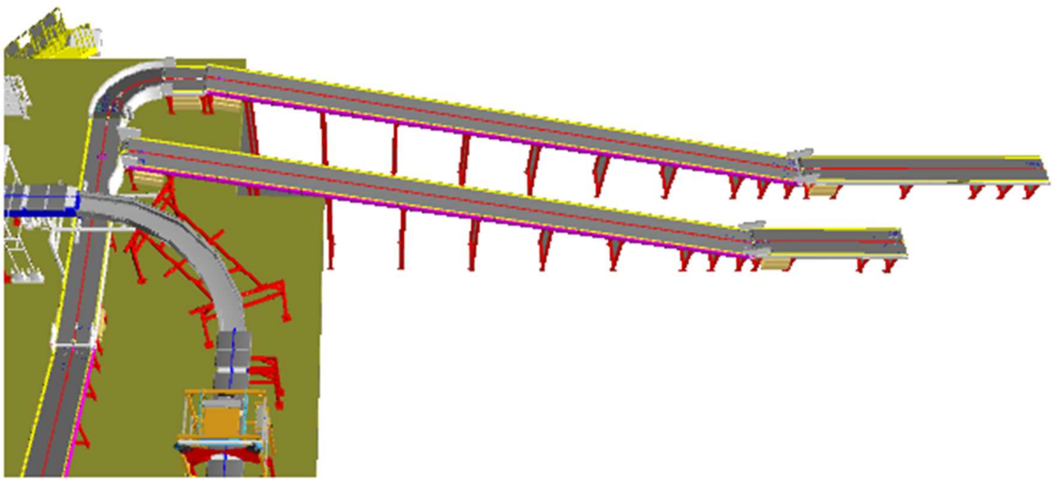
2- VDS Loop

6.1.1 Bulk Infeed Conveyor

Falcon's PVC Belt Conveyors are modular & robust in design. MS profile is used to build conveyor frame.

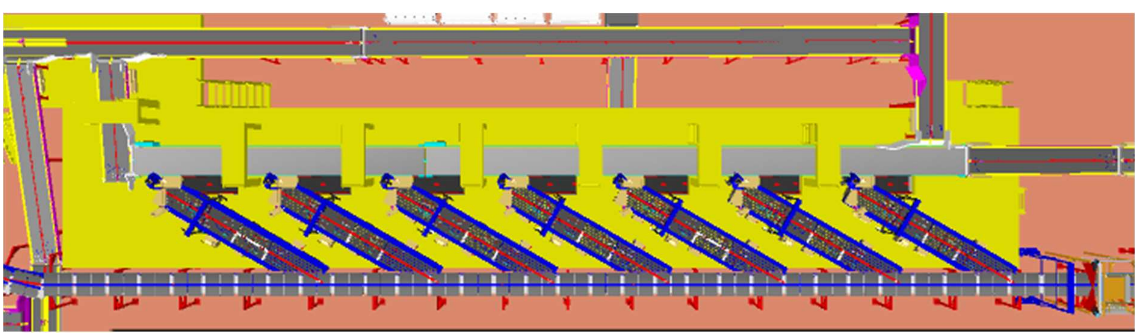
Some Silent Features of Falcon's Belt conveyor:

- Derco /Forbo Belts
- High Friction Belts for Inclines and Declines
- Side Guides



6.1.2 VDS Loop

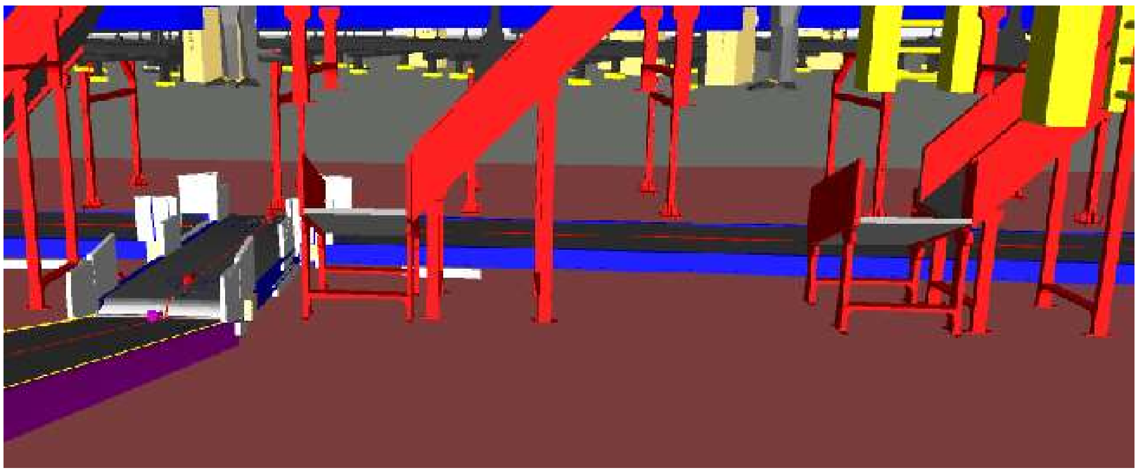
The loop Volume Distribution system enables seamless picking of incoming bulk flow of shipments from the infeed conveyor. The left-over shipments are re-circulated in the loop to repeat the process.



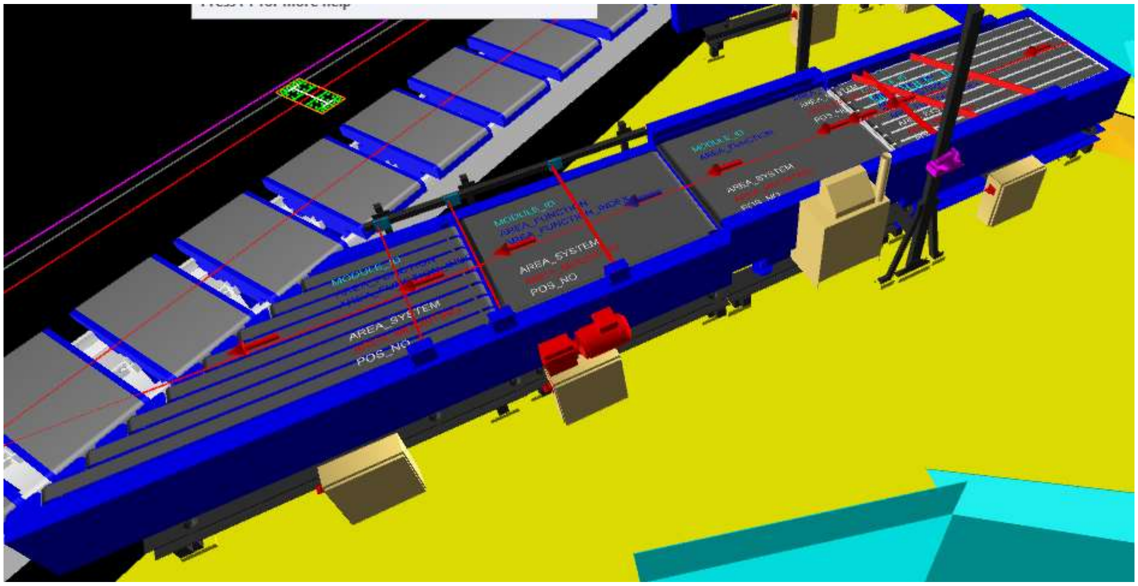
6.2 Irregular Shipments take out Conveyor

Provision of Chutes are provided at Induct Stations to Take out Odd Shaped Shipments from the System.
Operator identifies these shipments visually or by Size Gauge and drop them onto the chute and Shipments slides through the Chutes on the Conveyor.

The conveyor terminates at the non-con manual sorting station where they are manually sorted into PTL racks manually.



6.3 Feedlines



1. The Induct zone is equipped with 7 Induct lines.

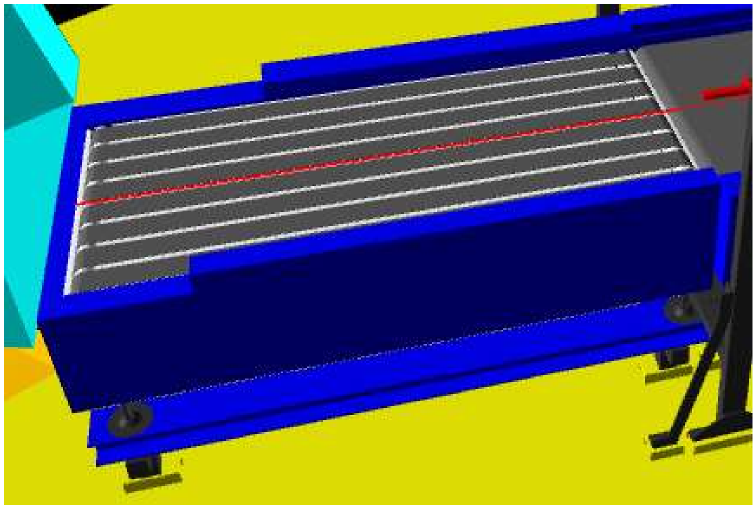
2. Each Feedline has four Conveyor Modules- 1. Loading & Orientation Conveyor, 1 weighing conveyor, 1 Intelligent feeding unit conveyor and 1 angle merge conveyor

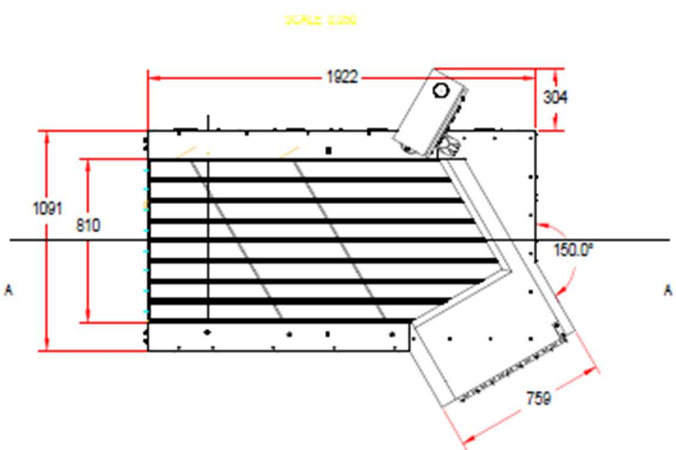
3. Sensors are placed on the Induct Lines to determine the exact position and dimensions (Length and Width) of the Parcel on the conveyor and prepare the Cross-Belt Cell for receiving the parcel with high precision.

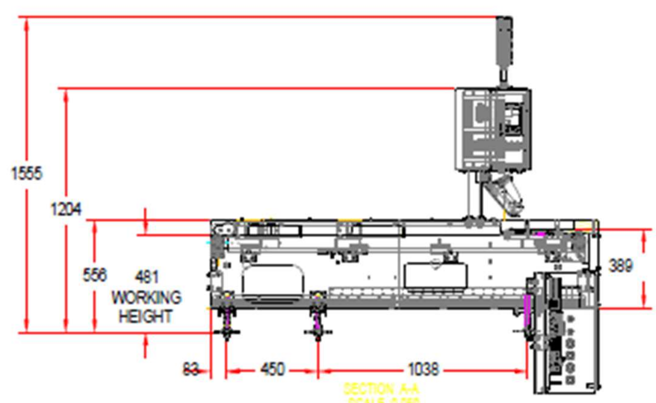
4. Over the feedline, parcels are evenly spaced and smoothly transferred to the main Loop.

6.3.1 Loading & Orientation conveyor

Parcels are loaded on this conveyor with proper orientation so that the barcode sticker is at the top. Orientation is achieved with the help of fixed aligner.





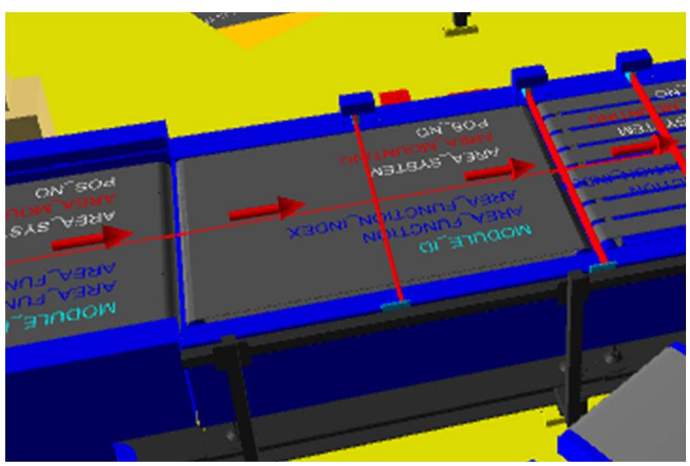


6.3.2 Weighing conveyor

This conveyor automatically measures the weight of the parcel.

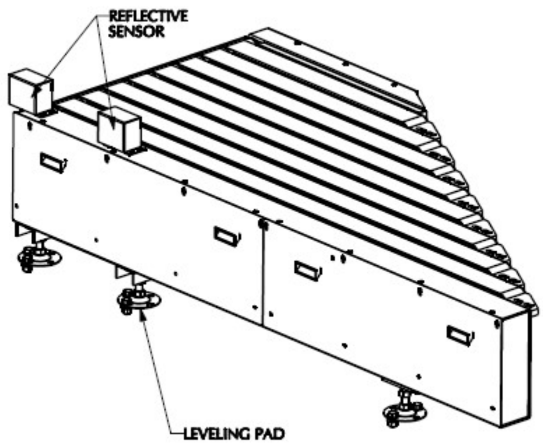
6.3.3 Intelligent Feeding Unit

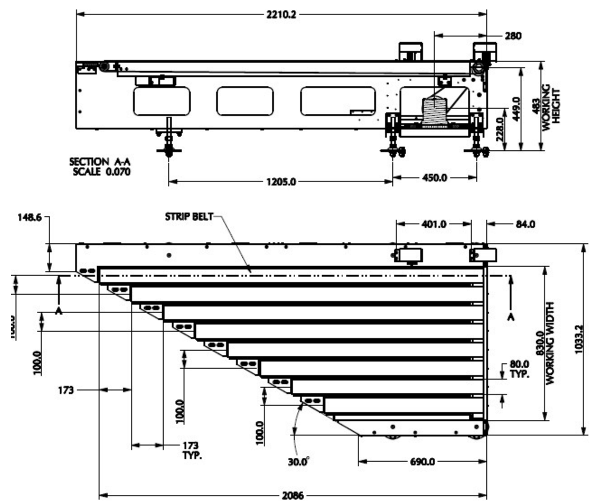
This conveyor is a variable speed special purpose module that creates space between parcels as well as regulates feeding on the intelligent merge.



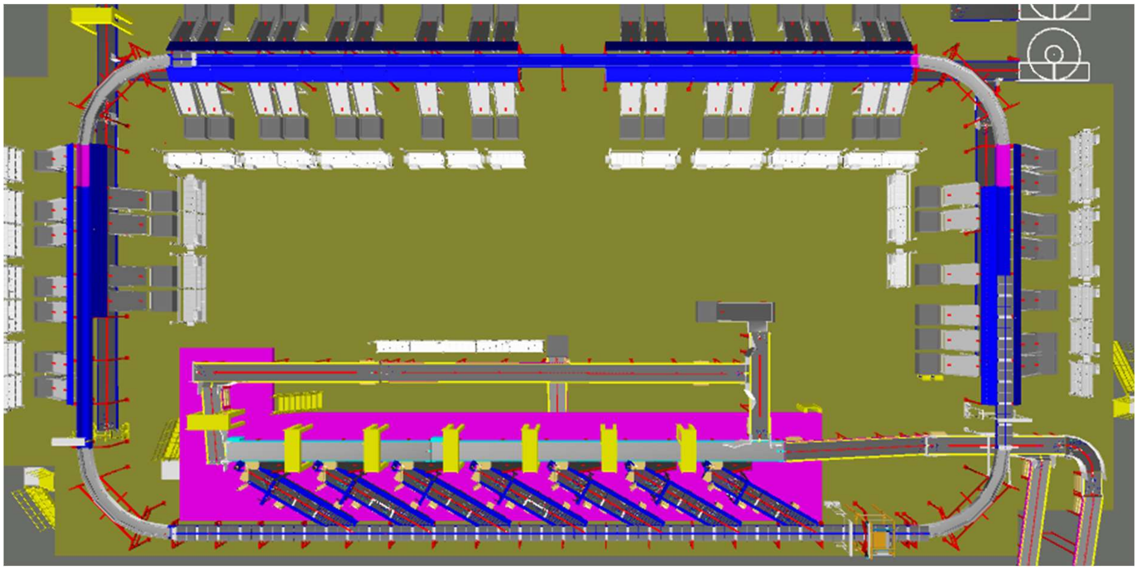
6.3.4 Intelligent Merge Conveyor

This is 30° triangular Dual Belt high-speed conveyor used for inducting parcel/boxes directly on to the sorter. The Belts are Strip Belts for smooth parcel movement.





6.4 Main Loop



Main Loop

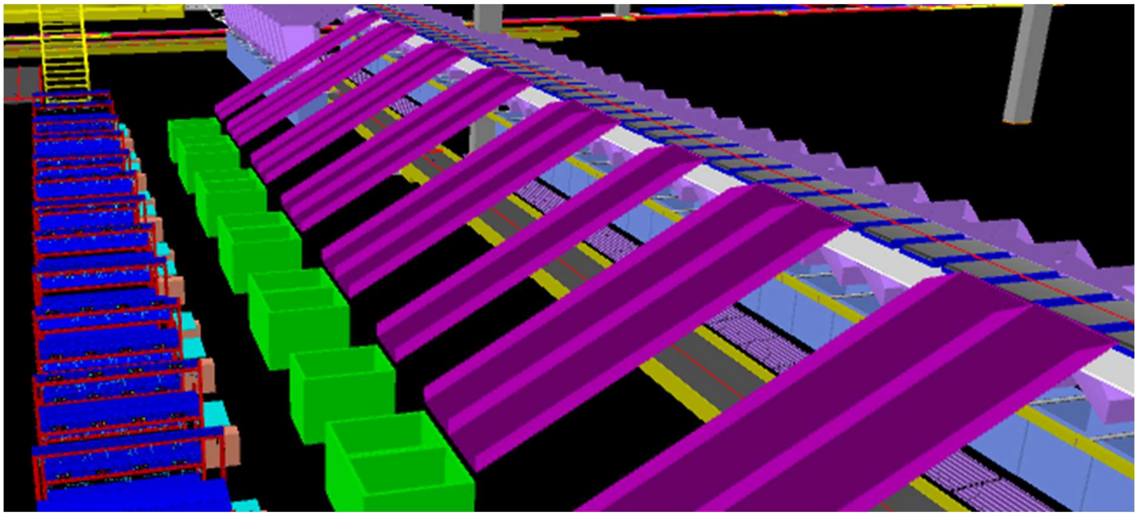
1. The proposed Sorter will be installed on the mezzanine (4000 mm clear height from ground). The Sorter will run at 2600 mm w.r.t. the mezzanine level at 2.1 m/s speed.

2. We have provided Dual Belt Carriers with 1175 mm pitch and Belt Size of 800x495 mm. Please note in dual belt carriers each individual belt can carry one parcel provided the parcel dimensions are within the single belt carriage's permissible limit. <u>Please note. dual belt features, i.e., the ability of the sorter to place over-sized products on two belts, has not been considered due to shortage of space and lack of utility for the parcel profile which Delhivery intends to run on the sorter.</u>

3. As Parcels pass through the Barcode Scanning Tunnel on the Loop, their barcodes are scanned, and chutes are assigned. Once the shipment reaches respective chute, carrier carrying the shipment for the chute, will actuate and drop the shipment in the assigned chute.

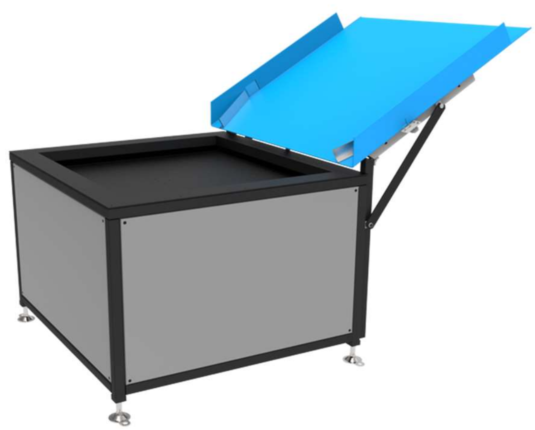
6.5 Output Chutes

Falcon designed Sliding type chutes



1. **PTL Chutes with spring loaded bins**

   These chutes drop the parcel in the bellow type spring loaded bins for secondary sorting by means of PTL racks



Chute IOs	Description
Chute Full Sensor	To alert the system that the chute is full and mark that chute unavailable for further sortation.
Tower Lamp	To indicate the operator action on a chute

6.6 PTL Racks & Modules

PTL Racks with PTL light modules has been provided for secondary sorting of parcels that have been dropped in the PTL chutes.

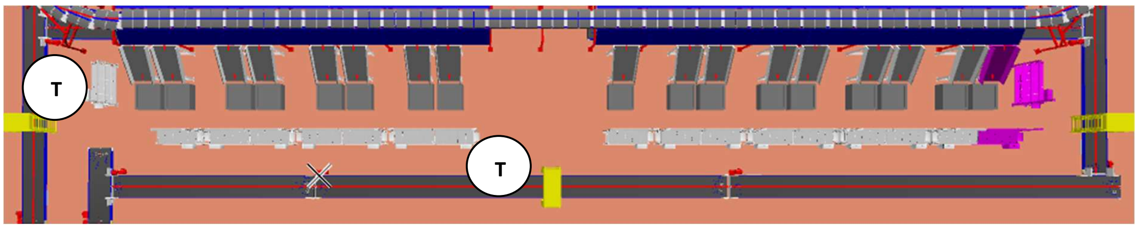
Rack Configuration: 3 Rows, 4 Columns, total of 12 modules/rack in one chute. There are 5 no's dual PTL chutes where there are 2 racks of total 24 PTL modules are mapped to a single chute.

- Operator will pick up a parcel from the PTL chute bin and scan the parcel barcode by handheld scanner
- The light of the particular block in the PTL rack which has been mapped to the destination location of the parcel will glow.
- Operator will place the parcel in the bag framed in that module and acknowledge by pressing the confirmation button on the module.
- This process will be repeated for all parcels falling in that chute bin.
- Once a bag is full, operator will take out the bag paste specific tag on bag seal and place it on the bag take-away conveyor.
- New bag will be placed in the empty module .
- Due to space constraint control boxes will be placed at the back side of the racks.
- However an extended plate of not less than 205 mm width will be provided at the side of the racks for placing bar code label printer(printers in Delhivery scope)

10.7 **Bag Take Out Conveyors**

For Direct Bagging Chutes, Conveyor Network is planned to Take out Order bags from the System.

Once the Bags are filled, they are pushed onto the Take-out Conveyors



T **Take Out Conveyors**

7. Proposed System Technical Details

7.1 Mechanical Equipment- Bill of Materials

Pos.	Qty.	Description	Value
FM1	**1**	**Conveyor Package** • Powered Belt Conveyors • Powered Turn Conveyors (90 Degrees) • Modular Conveyors	~295 Metres (28 Conveyors) 1 Nos. ~30 Metres (2 Nos). .
FM2	**7**	**Feedlines** Consists of • Single Operator Loading Conveyor. • Weighing System (Bizerba Make) • Dynamic Spacing System • Angle merge • Hand Scanners for Manual Scanning	1 Set 1 Set 1 Set 1 Set 1 No.
FM3	**1**	**Sorter** **1 Loop Cross Belt Sorter** • Sorter Height • Sorter Length • Sorter Speed • Sorter Drive • Carrier Pitch Including: • Standard Sorter Supports • Empty Carrier Detection System • Product Centring System • Camera based Barcode reader • Dimension Scanning System • E-Stops • Hooters • Fencing • Netting • Overshoot Assembly	2600 mm(from mezannine) ~143.5 m ~2.1 m/s Linear Motor 1175 mm
FM4	**1**	**Chutes** • PTL Chutes • Rejection Chutes • Irregular Chutes • Chute Full Sensors • Tower Lights • Contingency Chute • Spring Loaded Fixed Bin	55 Nos 3 Nos 7 Nos 59 Nos 59 Nos 1 Nos 60 Nos

FM5	**1**	**PTL** • PTL Racks • PTL Modules • PTL Control Box • PTL Barcode Hand scanner 1-D	708 Nos 660 Nos 55 Nos 55 Nos
FM6	**1**	**Steel Works** • Operator and Maintenance Platform **<u>Please note- The current Proposal does not include the Price of Steel Works and the same shall be discussed separately with yourself.</u>**	~2540 SQM

7.2 Electrical Equipment

Electricals			
FE1	**1**	**Consists of** • Main Power Distribution panels • Main Control Panels • Feedline Control Panels • Conveyor Drive Panels • Sorter Drive Panels • Network Switches • Field Cabling • Cable trays • Remote I/O Panels	Included

7.3 Control System

Components			
FC1	**1**	**Controls and IT Infrastructure Controls** • Weigh Scale Control • Barcode Scanner Controls • Dimension Scanner Controls • Conveyor Package Controls • Sorter Control • Induct Control • Smart Chutes Control **IT** • Control IT Software Package • High Availability Server Setup (Microsoft Clustering based) • Networking Switches	Included

7.3.1 Technical specifications of Sorter

Items	Make
Sorter Carrier Type	CBS Dual Belt
Sorter Speed in m/s	Up to 2.1 m/s
Sorter Loop Length	~143 m
Length of the Sorter	Layout Attached
Width of the Sorter including Chutes	Layout Attached
Sorter Actuation Technology	Electric
Noise Level	<75 Db
Chutes	
Chutes + Rejection	55+3
Height of Sorter	~2600 mm w.r.t. Mez
Track Material & Type of Design to move the Carrier trolleys	~Steel Track
Carrier Design	
Carrier Motor Make	Falcon
Carrier Pitch	1175
No. of Carriers	122
Drives	
Motor/Drive type	Linear Induction Motor
Power Consumption	Will be specified during DAP
Induction Mechanism to the Sorter	
Automatic Feedlines	7
No. of Barcode Scanners	1
Power and Data Transmission	
Power Transmission	Over Continuous Bus Bar
Parcel Positioning System	1
Parcel Centering System	1
Empty Carrier Detection System (Camera Based)	1
Maintenance Speed of the Sorter- Jog Mode	0.8 m/s
Area Requirement (L x B)	Layout Attached

8. Key Components Make

Items	Make
Belts	Forbo/Derco
Rollers	Falcon
Cross Belt Carriers	Falcon
Linear Induction Motors	Falcon
Feed Line Motors	Induction Motors – SEW/ Lenze
Barcode Readers	SICK / Cognex
Wireless Barcode Scanner	Honeywell/Motorola
Load Cells	MT/Bizerba
Volume Scanners	SICK
Encoders	SICK
Sensors	Sick/Leuze/ P&F
PLC	Siemens's
Control Panels	Rittal/ BCH
MDR Rollers	Falcon/ Pulse/ Interrol
Power Transmission Systems	Vahle

9. Warranty Period

Falcons offered System comes with a standard warranty of 12 months from successful commissioning of the system or 15 months from the date of completion of delivery, whichever is earlier. In case of delay in lifting the material by client, the 15 months warranty period will start on the 15th day after material readiness intimation is formally communicated to client

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- 24 X 7 Telephonic, Email and Remote Service Support when required
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles continued:
  - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
  - Negligence or abnormal use of equipment.
  - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
  - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

10. Exclusions

The scope of supply includes all parts which are defined in the Supplier’s quotation.

All other parts which are not defined in the Supplier’s quotation do not belong to the Supplier’s scope of supply and are excluded. The following parts are also excluded:

- **Air Compressor**
- **Construction Power**
- **Steel Works- All steel works including the sorter mezzanine as well as operator platforms at the induct zones and maintenance platforms for o/h conveyors will be in Delhivery scope.**
- Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- Electrical power supply and wiring to the main control cabinets
- UPS for Controls and Drives
- Electrical Power for Installation
- Network Cabling up to the Main Server Rack
- Intermediate wiring to parts which are to be supplied by the Purchaser/others
- Emergency/Uninterruptable power supply
- Fire-alarm and fire protection devices
- Traffic and route markings
- Laydown Area
- Ram protection devices
- Cat walks, bridges, maintenance aisles and platforms
- Assembly tools like forklifts and hoisting machines
- All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- Any Kind of Civil work
- Any adjustment of the Supplier’s scope of supply to local rules and regulations
- X-Ray machines
- Roller cages/ Pallets
- Simulation and 3D animation of the sorter system
- Interface with other equipment not specified in this offer.
- Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- The supply and installation of fencing around the different corridors.
- Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.

11.Commercial Offer & Terms and Conditions

11.1.1 Itemised Cost of Sorting System

Heads	Sub heads	Qty	Cost
Conveyors	Powered Belt conveyor	144	₹ 71,26,394
	Turns	1	₹ 7,70,000
	Modular conveyor	30	₹ 23,63,533
Sorter	Inducts	7	₹ 77,61,635
	Loop	144	₹ 3,90,26,490
	VMS	1	
	ICR	1	
Weighing conveyor		7	₹ 47,62,590
Bar code scanner for semi Auto		7	₹ 1,26,000
Chutes	Sliding Chutes+sensor+I/O+Tower Lamps	66	₹ 29,89,420
	Spring loaded Bins	60	₹ 23,10,000
PTL System	Hand held scanners	55	₹ 76,43,920
	Modules	708	
	Frames	660	
	Control Boxes	55	
	Rejection Frames		
Bag Conveyors		151	₹ 68,80,569
Misc	Tool Kit	1	₹ 79,800
Controls & IT		1	₹ 26,55,574
Project Management		1	₹ 4,51,983
Packing & forwarding		1	₹ 16,85,919
Installation & Commissioning		1	₹ 42,14,796
Total			₹ 9,08,48,623
BCA			4.5%
Nett Price			₹ 8,67,60,435
Old Discount			1%
Final Price			**₹ 8,58,92,831**

11.1.2 Price Terms

The Total Price is to be understood.

- Ex Works
- Taxes: Extra as applicable
- Excluding Freight & Unloading & MHE's as required.
- Including Server PC
- Including Packing as described
- Excluding Steelwork i.e. (Mezannine , Operator Platform and Stairs)
- Including Installation Supervision Cost & Commissioning.
- Scaffoldings, Man Lift etc. required for reaching the maintenance platforms shall be in customer scope.

*Material Unloading & Laydown area is in customer scope.

11.1.3 Payment Schedule

The Supplier Quotation is based on the following payment schedule:

Payment %age	Project Stage
40%	Advance along with PO
50%	With 100% taxes within 30 days of material reaching at Site
10%	After Installation & Commissioning

(*): *All invoices (if dues as per payment terms) will be cleared with in 30days from the date of receipt of undisputed invoice by the buyer.*

11.1.4 Schedule for Delivery and Installation

The complete system is to be delivered, installed, and commissioned by 30th Sep 2023.

* *Detailed Project Plan will be shared during DAP.*